

KOHLER AI Bathroom Intelligence

Research, Data & System Workflows

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1. User Workflow — Main Product

This is the workflow the judge should experience.

LANDING



Start a project



UPLOAD BATHROOM IMAGE + ENTER / CONFIRM DIMENSIONS



AI ANALYZES SPACE

- walls
- doors
- windows/openings
- existing fixtures
- approximate dimensions
- spatial constraints



USER CONFIRMS / CORRECTS



DESIGN PREFERENCES

- Budget
- Aesthetic
- Fixture finish
- Mood intensity



COLOUR + STYLE PROFILE



KOHLER PRODUCT RETRIEVAL



HARD CONSTRAINT FILTER

- budget
- category
- dimensions
- installation
- product dependencies



COMPATIBILITY CHECK

- requires
- included_component
- order_with



SPATIAL VALIDATION

- footprint

- clearance
- collision
- door swing
- openings
- mounting

↓

MULTI-OBJECTIVE OPTIMIZATION

- space
- budget
- style
- colour harmony
- functionality
- sustainability where documented

↓

TOP 2-3 DESIGNS

↓

AI EXPLANATION

↓

WHAT-IF EXPLORATION

↓

FINAL DESIGN

↓

DESIGN BRIEF / DEALER HANDOFF

2. Research Workflow

This was the foundation before coding.

Problem identification

↓

Bathroom planning pain points

↓

Existing digital solutions

↓

Competitor / market research

↓

KOHLER ecosystem research

↓

Identify gaps

↓

Define opportunity

↓

Define product concept

↓

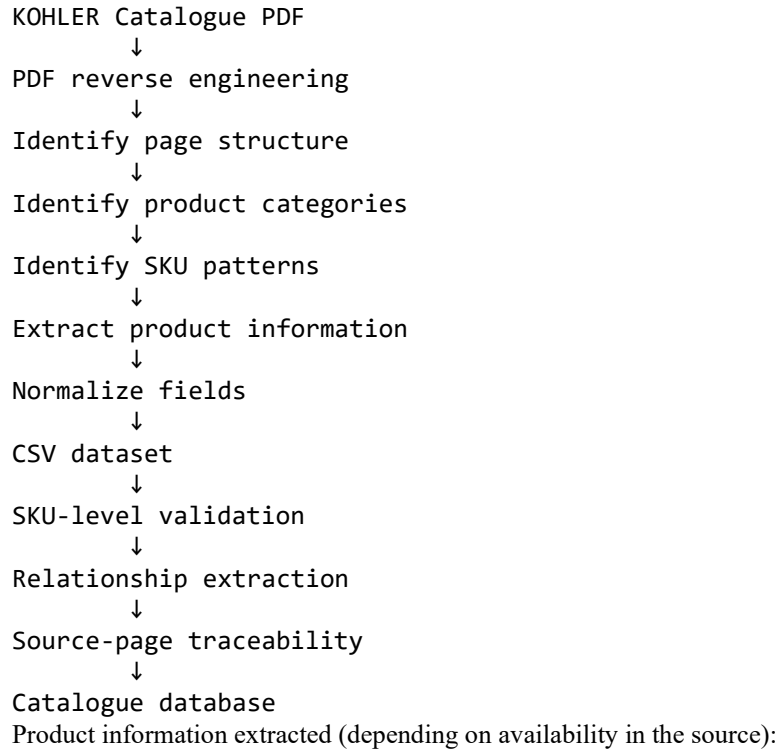
Define technical requirements

The key idea we arrived at was that the system should not behave like a generic chatbot. It should behave more like:

A digital KOHLER showroom consultation backed by deterministic engineering.

3. KOHLER Catalogue Workflow

This is particularly important because the catalogue is the source of product truth.



- SKU
- Product name
- Category
- Subcategory
- Domain
- Price
- Dimensions
- Finish
- Finish code
- Material
- Installation type
- Configuration
- Components
- Requirements
- Included components
- Order-with relationships
- Specifications
- Source page
- Source document

The important architectural principle was:

The AI does not invent catalogue facts. The catalogue database is the source of truth.

5. Lookbook Intelligence Workflow

The lookbook was treated differently from the product catalogue.

KOHLER Lookbook
↓
Page classification
↓
Bathroom / kitchen scene identification
↓
Design style extraction
↓
Material identification
↓
Colour relationships
↓
Fixture / product visual cues
↓
Spatial composition
↓
Design patterns
↓

Structured lookbook dataset

Lookbook intelligence

We looked for information such as:

- design style
- room type
- colour relationships
- materials
- fixture finishes
- visual composition
- spatial arrangement
- product/finish combinations
- design cues

The lookbook therefore acts as design intelligence, not product truth.

6. Colour Intelligence Workflow

This became a separate intelligence layer.

External colour references
↓
Extract colour combinations
↓
Preserve HEX values
↓
Identify colour roles
↓
Analyse harmony
↓
Analyse temperature
↓
Analyse saturation
↓
Analyse contrast
↓

Map to design themes
↓
Rank usable palettes
↓
Colour intelligence dataset

Then:

User preferences
↓
Theme / aesthetic
↓
Mood intensity
↓
Fixture finish
↓
Colour intelligence
↓
Recommended palette
↓

Compatible KOHLER finishes

And we deliberately maintained the distinction:

Reference colour ≠ KOHLER catalogue finish. The colour dataset provides design guidance; actual KOHLER finishes must come from the catalogue.

7. Image Processing Workflow

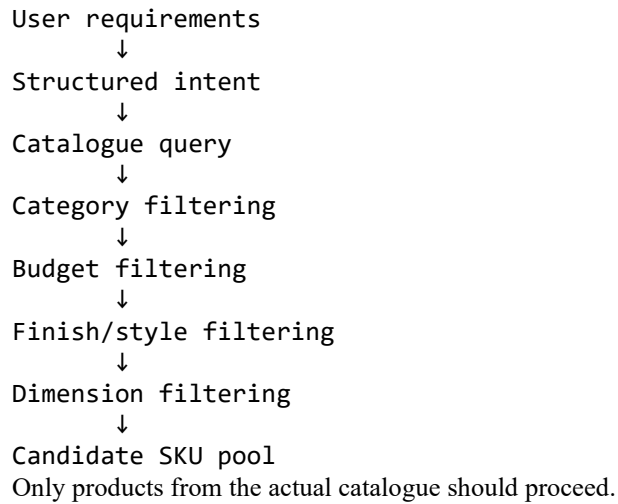
The user's image isn't treated as decoration. It becomes an input to spatial reasoning.

Bathroom photograph
↓
Image analysis
↓
Detect visible geometry
↓
Identify fixed obstacles
↓
Identify doors/windows
↓
Identify existing fixtures
↓
Estimate spatial dimensions
↓
Generate room representation
↓
Confidence levels
↓
User confirmation/correction
↓
Spatial constraint engine
The important safety/accuracy principle:

Image-estimated measurements are treated as estimates until confirmed.

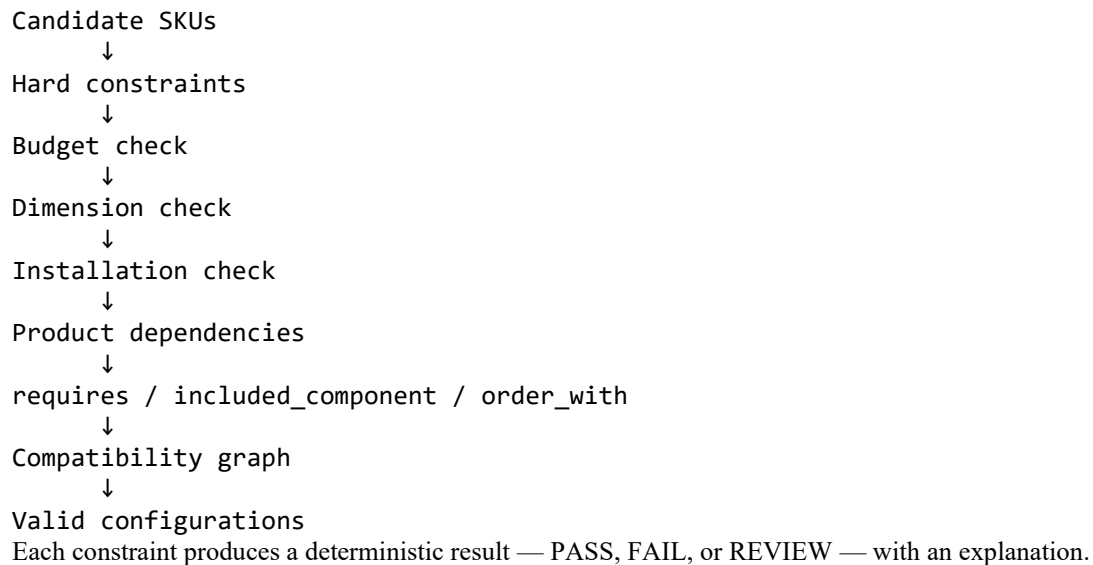
8. Retrieval Workflow

Once the system understands what the user wants:



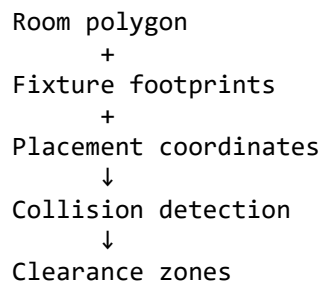
9. Constraint & Compatibility Workflow

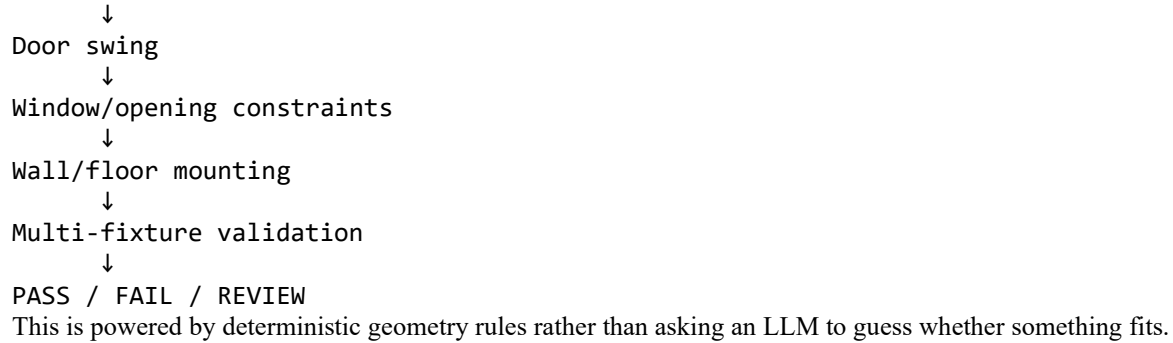
This is where the project becomes more than a recommendation system.



10. Spatial Workflow

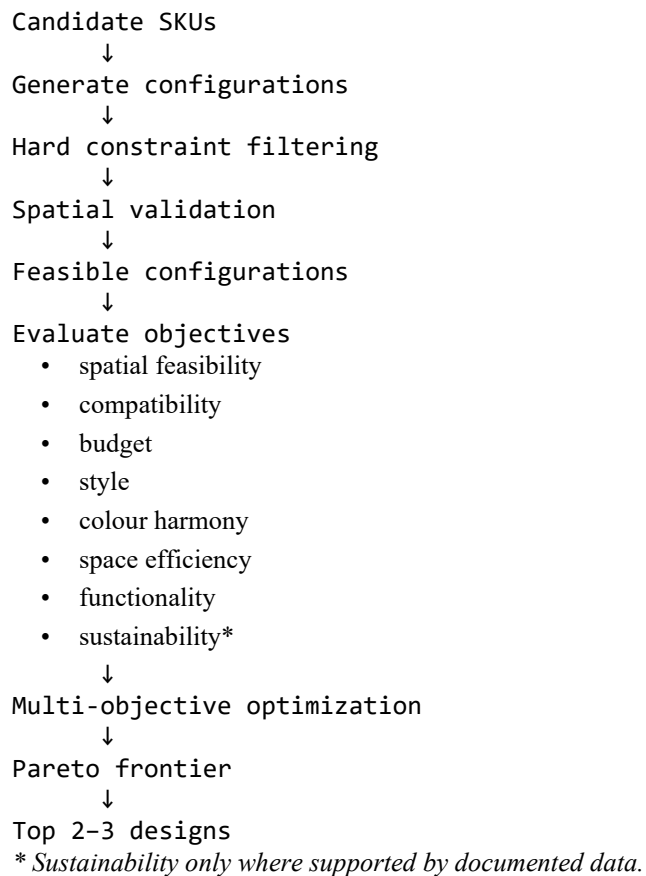
The spatial engine answers: Can these actual KOHLER products physically coexist in this bathroom?





11. Optimization Workflow

Instead of returning the first valid product combination:



17. Overall Research → Product Workflow

The master diagram for the presentation/documentation:

